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Volume 1

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Introduction

The aims and objectives of the unit

Computer science depends upon the science of mathematics and without the mathematical ideas that underpin it, none of the marvels of modern computer technology would be possible. In order to study for a degree in Computing and Information Systems, you need to understand and feel easy with some essential mathematical ideas. The topics in this course have been selected with that in mind and you will find that you use many of the ideas and skills introduced in this unit directly or indirectly in the other units in this degree programme. You will also gain experience of the way that mathematicians and computer scientists express their ideas using symbols and make their statements precise.

Although this unit does not go very deeply into any one topic, we hope that you will gain sufficient confidence from studying it to consult mathematical textbooks to pursue areas that particularly interest you or about which you need more information. Currently, there are two optional half units available at Level 3 of this degree programme that develop aspects of this unit directly.

Using this subject guide effectively

The subject guide for 2910102 [CIS102] is in two volumes, and the material is presented in a convenient order of study. Taken together, the two volumes contain a complete account of the examinable topics in the unit. The chapters are not all of equal length and the time you should allow for studying them will depend very much on your previous mathematical experience.

It is very important to understand that you only learn mathematics by *doing* it. Ideas and methods that may seem complicated at first sight will become more familiar and natural with practice. You will then be able to apply the ideas you learn in this course to your other units with more facility, and will find the exam questions easier to answer. The exercises at the ends of each chapter are therefore a crucially important part of the course and we strongly recommend you to try all of them. Answers to the exercises in this volume are included as an Appendix. You can get extra practice, particularly on topics that you find difficult, by trying additional questions from an appropriate textbook.

Textbooks

Although this subject guide gives a complete account of the course, we do encourage you to consult a textbook for alternative explanations, extra examples and exercises, and supplementary material. There are a number of books on Discrete Mathematics available. Many of them cover most (but not necessarily all) of the topics in the the syllabus. They vary in style and the level of mathematical maturity and expertise they presuppose on the part of the reader. We recommend that you obtain a copy of one of the following titles. References are given to them where appropriate in the subject guide.

Either

Susanna S. Epp, *Discrete Mathematics with Applications.* 2nd Edition (Brooks/Cole 1995) [ISBN 0-534-94446-9(hbk)]

or

Molluzzo, J. C. and Buckley, F. *A First Course in Discrete Mathematics.* (Waveland Press, reprinted 1997) [ISBN 0-88133-940-7]. Referred to as M&B throughout the text. Available in the USA only.

A list of other suitable textbooks known to the author of this guide and in print at the time of writing is included in this volume as an Appendix.

Discrete Mathematics is a comparatively modern area of study and the notation in some topics has not been completely standardised. This means that you may find occasional differences between the symbols and terminology used in textbooks by different authors. The examination papers will follow the usage in this subject guide, however.

Assessment

Important: the information and advice given in the following section is based on the examination structure used at the time this guide was written. However, the university can alter the format, style or requirements of an examination paper without notice. Because of this, we strongly advise you to check the rubric/instructions on the paper when you actually sit the examination. You should also read the examiner's report from the previous year for advice on this.

This unit is examined by a three hour written paper. Currently, ten questions are set and full marks for the paper are awarded for complete answers to *all* of them. Each question carries the same number of marks. In general, there is at least one question on the material in each chapter of the guide, but some questions may require knowledge or techniques from more than one chapter.

You may answer the questions in any order. If you feel nervous, it may help you to start with a question on a topic you feel really confident about. For the most part, the questions are very similar to worked examples or exercises in the subject guide, so that any student who has understood the material and revised thoroughly for the exam should be able to answer most of them quite easily. Some parts of questions may contain a new twist that requires more careful thought. If you get stuck on part of a question, do not spend *too* long trying to figure it out, because this may mean that you do not have time to attempt another question that you could answer quite easily. Leave a space in your script and come back to the bit you found difficult when you have attempted as much as you can of the rest of the paper. Some questions may also ask you for a definition or the statment of a result proved in the subject guide. These need to be carefully learnt. You may express a definition in your own words, but make sure that your words cover all the points in the definition in this guide. A detailed marking scheme is given on the paper and that can be used as a guide to the length of answer expected. If there is just 1 or 2 marks awarded for a definition, for example, only one sentence is expected, not a half page discussion.

Although most students find that attempting past examination papers is reassuring, you are strongly advised not to spend *all* your revision time in this way, because every year the questions will be different! It is much better to revise thoroughly the ideas and skills taught in each chapter until you are quite confident that you really *understand* the material. When you have finished your revision, test yourself by answering the specimen examination questions to time. Suggested solutions are also included, so that you can check your answers and see the level of detail required.

In conclusion, we hope you will enjoy studying this unit and find the material interesting and challenging, as well as increasing your understanding and appreciation of your other units.

Chapter 1

Numbers Systems

Summary

Number bases; the decimal, binary and hexadecimal number systems; calculating in the binary and hexadecimal systems; converting between number bases; factors, multiples, primes; rational numbers and their decimal representation; irrational numbers; inequalities; floating point notation.

References: Epp Sections 1.5 (pp 57-61, 70-73), 3.2 (pp 125-127) *or* M&B Sections 1.1, 1.2, 1.4.

Introduction

The history of mathematics began thousands of years ago with numbers and counting. As well as the practical importance of numbers in everyday life over the intervening centuries, the properties of numbers have fascinated mathematicians and laymen alike and been a focus for the development of deep mathematical theories. Today, numbers are more important in our lives than ever before, with information in every field, from pin numbers and barcodes to music and television transmissions recorded digitally. In this chapter, we consider various ways in which numbers can be represented.

1.1 Number Bases

Learning Objectives

After studying this section, you should be able to:

- convert a numeral given in any base to a decimal numeral;
- demonstrate understanding of the binary number system by performing simple arithmetical calculations in the binary system;
- convert a binary numeral to hexadecimal and vice-versa;
- express a decimal numeral in the binary and hexadecimal number systems.

1.1.1 Numbers in base 10

Let us start by reminding ourselves how a string of digits is used to represent a whole number in our familiar number system. Consider the number 73589. We can think of each digit as appearing in a labelled box, where each box has a **place value**, according to its position, as illustrated below.

$\cdots$	10^4	10^3	10^2	10^1	1
$\cdots$	7	3	5	8	9

The convention that the *place value* of each digit determines how much it contributes to the value of the number allows us to represent any whole number, however large, using only the ten different

digits $0, 1, 2, ..., 9$. The name *decimal*, coming from the Latin word for *ten*, reflects the fact that every place value in our number system is a power of 10. Note that the right hand *units* box also has a place value which is a power of 10, since $1 = 10^0$, but we shall represent this place value by 1 for convenience.

Definition 1.1 *Because each place value in the decimal system is a power of 10, we say that the decimal system has* **base 10**.

For the moment, we restrict our discussion to the *integers*, that is, whole numbers. We shall see later how the system of place value can be extended to fractional numbers by using *negative* powers of the base.

Instead of using boxes, we can write out the number 73589, for example, in **expanded notation**, as

$$7(10^4) + 3(10^3) + 5(10^2) + 8(10^1) + 9(1).$$

Example 1.1 Suppose that $a_3a_2a_1a_0$ represents a decimal number, where the symbols a_3, a_2, a_1, a_0 each represent one of the digits $0, 1, 2, \ldots, 9$ (and are not necessarily different from one another). Then we can write $a_3a_2a_1a_0$ in expanded notation in base 10, as

$$a_3(10^3) + a_2(10^2) + a_1(10^1) + a_0(1) \ \square$$

Numbers in base 5

The decimal number system seems so natural and familiar that it is hard to realise that its use is entirely fortuitous, due presumably to the accident of our having ten fingers to count on. Other peoples and civilizations have based their number systems on other integers, such as 5, 12 or 20. The Babylonians based theirs on 60 and it is to them that we owe the division of both an hour and a degree of turn into 60 minutes and each minute into 60 seconds. There are vestiges of counting systems based on 12 and 20 in the old British systems of weights and measurement and in the special words *dozen* for 12, *gross* for 144 (or 12^2) and *score* for 20 that have remained in the language.

Let us consider first how numbers are represented in a number system based on 5 instead of 10. The digits are just $0, 1, 2, 3, 4$ and these are the *only* symbols we are allowed to use to record a number. The place value of each digit is a power of 5, so that we count in units, 5's, 25's and so on.

A natural way to think of the system in base 5 is that we are using the fingers of one hand only to count with, and 5 units makes a "hand". You may also have met a system of counting in 5's to keep a *tally* when making a count for a statistical survey. Thus to represent the decimal number 13, we would need 2 "hands" + 3 "fingers", which we record as $(23)_5$ in base 5. Working in this way, we obtain the table below, which records the base 5 representation of the numbers from 1 to 16. Note that when we write a number in a base other than 10, we enclose it in brackets and append the base as a suffix. Unless otherwise stated, numbers with no suffix are in base 10.

base 10	*base* 5	*base* 10	*base* 5
1	$(1)_5$	9	$(14)_5$
2	$(2)_5$	10	$(20)_5$
3	$(3)_5$	11	$(21)_5$
4	$(4)_5$	12	$(22)_5$
5	$(10)_5$	13	$(23)_5$
6	$(11)_5$	14	$(24)_5$
7	$(12)_5$	15	$(30)_5$
8	$(13)_5$	16	$(31)_5$

To convert a number from base 5 to decimal, we can use the box system.

Example 1.2 To convert the base 5 number $(3214)_5$ to decimal, we enter the digits in the boxes as shown below.

5^3	5^2	5^1	1
3	2	1	4

Thus, using expanded notation, $(3214)_5$ represents

$$3(5^3) + 2(5^2) + 1(5) + 4(1) = 434. \square$$

1.1.2 The binary system

The electronic circuitry of a computer easily lends itself to the representation of numbers using a two digit number system. This is because a digital computer is really just a gigantic collection of on/off switches, with an intricate set of connections between them. Thus it is natural to represent the two positions of a switch by the digits 0 and 1.

Definition 1.2 *The number system with base 2, having just the two digits 0 and 1, is called the* **binary** *number system. Each binary digit is called a* **bit** *(short for* **binary digit***).*

A string of bits, like 10001011, for example, is called a **binary string**. The number of bits in the string is called the **length** of the string. A binary string of length 8 is called a **byte**. The computer uses bytes as the basic unit of information. We shall see later that since there are two choices, 0 or 1, for each of the eight bits in a byte, there are altogether $2^8 = 256$ different bytes.

We shall now consider how to represent numbers in the binary system. The box system in binary giving the place value of each bit is shown below.

$\cdots$	2^5	2^4	2^3	2^2	2^1	1

The following table shows the binary equivalent of each of the decimal numbers from 1 to 16.

base 10	*base* 2	*base* 10	*base* 2
1	$(1)_2$	9	$(1001)_2$
2	$(10)_2$	10	$(1010)_2$
3	$(11)_2$	11	$(1011)_2$
4	$(100)_2$	12	$(1100)_2$
5	$(101)_2$	13	$(1101)_2$
6	$(110)_2$	14	$(1110)_2$
7	$(111)_2$	15	$(1111)_2$
8	$(1000)_2$	16	$(10000)_2$

Example 1.3 To convert the binary integer $(1001011)_2$ to the decimal system, we write it in expanded notation. You may find it helpful to put it into boxes first, as shown below.

2^6	2^5	2^4	2^3	2^2	2^1	1
1	0	0	1	0	1	1

Thus we have

$$\begin{aligned}(1001011)_2 &= 1(2^6) + 0(2^5) + 0(2^4) + 1(2^3) + 0(2^2) + 1(2^1) + 1(1) \\ &= 2^6 + 2^3 + 2^1 + 1 = 75. \square\end{aligned}$$

In practice, when finding the decimal equivalent of a binary number, it is easiest to work from *right to left*. Thus in the previous example, we would start from the units and compute successively:

$$\begin{aligned}1 + 1(2) &= 3 \\ &+ 0(4) = 3 \\ &\quad + 1(8) = 11 \\ &\quad\quad + 0(16) = 11 \\ &\quad\quad\quad + 0(32) = 11 \\ &\quad\quad\quad\quad + 1(64) = 75.\end{aligned}$$

With a little practice, you will be able to do this directly from the binary form of the number without using expanded notation.

Example 1.4 Consider the binary number $(a_4a_3a_2a_1a_0)_2$, where each of the symbols a_i represents a bit (0 or 1). We can write this in expanded notation as

$$a_4(2^4) + a_3(2^3) + a_2(2^2) + a_1(2^1) + a_0(1). \square$$

1.1.3 Calculating in the binary system

In this section we shall drop the suffices on the binary numbers *in the body of the calculations.* Since these will only involve numbers in base 2, no confusion will arise.

Binary addition

First, recall that when we add a column of digits in the decimal system that sum to a number greater than 9, we record the units in the column we are working in and carry the number of 10's into the next column to the left. When a column sums to more than 99, we carry the number of hundreds into the column two places to the left, and the number of tens into the column next to the left.

We procede in a very similar way in binary. To obtain the rules for carrying, suppose the column to be added is as follows (with any zeros omitted):

1. $(1)_2 + (1)_2$. This gives 2, or $(10)_2$ in binary. We record the 0 in the current column and carry a 1 to the next column to the left;
2. $(1)_2 + (1)_2 + (1)_2$, giving $(11)_2$. We record 1 in the current column and carry a 1 to the next column to the left;
3. $(1)_2 + (1)_2 + (1)_2 + (1)_2$, giving $(100)_2$. We record 0 in the current column and carry a 1 to the column *two places* to the left;

and so on.

Example 1.5 We sum the binary numbers $(1011)_2 + (1111)_2 + (11)_2$. In the format below, the numbers we carry at each step are recorded above the horizontal line.

Step 1

$$\begin{array}{cccc} & & 1 & \\ \hline 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ \hline\hline & & & 1 \end{array}$$

Adding the right hand column, we obtain $(11)_2$, and so we record 1 and carry 1 into the next column (that is, the 2^1 column).

Step 2

$$\begin{array}{cccc} 1 & & 1 & \\ \hline 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ & & 1 & 1 \\ \hline\hline & & 0 & 1 \end{array}$$

Adding the 2^1 column, we obtain $(100)_2$, and so we record 0 and carry 1 into the column two places to the left (that is, the 2^3 column).

Step 3

$$\begin{array}{ccccc} 1 & 1 & & 1 & \\ \hline & 1 & 0 & 1 & 1 \\ & 1 & 1 & 1 & 1 \\ & & & 1 & 1 \\ \hline\hline 1 & 1 & 1 & 0 & 1 \end{array}$$

Adding the 2^2 column, we obtain $(1)_2$ and so this time there is nothing to carry. Adding the 2^3 column, we obtain $(11)_2$ and so we record 1 and carry 1 into the 2^4 column. Adding the 2^4 column, we obtain $(1)_2$ and the process terminates.

Thus we obtain $(1011)_2+(1111)_2+(11)_2 = (11101)_2$. In decimal numbers, this addition is 11+15+3, and the sum we obtained is 29. □

Binary subtraction

Let us first revise how we deal with "borrowing" in base 10.

Example 1.6 We perform the subtraction $4003 - 597$.

$$\begin{array}{cccc} 3 & 9 & 9 & 13 \\ \hline 4 & 0 & 0 & 3 \\ & 5 & 9 & 7 \\ \hline\hline 3 & 4 & 0 & 6 \end{array}$$

Since 7 is greater than 3, we must borrow a 10. In this case, we do this from the 10^3 column. We rewrite the borrowed 1000 as $990 + 10$. So we reduce the entry in the 10^3 column from 4 to 3, replace the zeros by 9's in the next two columns and replace 3 by $10 + 3$ in the units column. The resulting row is recorded above the horizontal line. We can now obtain the answer by subtracting 597 from the row above the horizontal line. □

We shall now carry out the same process in binary, where we have to borrow a 2 when we have to perform the subtraction $(0)_2 - (1)_2$ in any column. The $(0)_2$ then becomes $(10)_2$, and we have

$$(10)_2 - (1)_2 = (1)_2.$$

This process is illustrated in the example below.

Example 1.7 We subtract $(1101000)_2 - (101011)_2$.

Step 1

$$\begin{array}{ccccccc} 1 & 1 & 0 & 0 & 1 & 1 & 10 \\ \hline 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline\hline \end{array}$$

As in the previous example, we have to borrow to perform the subtraction in the units column. The first column with a non-zero entry is the 2^3 column. We borrow $(1000)_2$ from this column and rewrite it as $(110)_2 + (10)_2$ (effectively, we are borrowing an 8 and rewriting it as $6 + 2$). The revised row is recorded above the horizontal line.

Step 2 We replace the row representing the number we are subtracting *from* by the row above the horizontal line at the end of Step 1. We can now perform the subtraction in the first three columns as shown below. When we reach the 2^3 column, we have to borrow again. We repeat the procedure described in Step 1. The revised top row is shown above the horizontal line.

$$\begin{array}{ccccccc} 1 & 0 & 1 & 10 & & & \\ \hline 1 & 1 & 0 & 0 & 1 & 1 & 10 \\ & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline\hline & & & & 1 & 0 & 1 \end{array}$$

Step 3 We replace the row representing the number we are subtracting *from* by the row above the horizontal line and perform the subtraction in the 2^3 and 2^4 columns. When we reach the 2^5 column, we have to borrow from the next column. The revised top row is shown above the horizontal line.

$$\begin{array}{ccccccc} 0 & 10 & & & & & \\ \hline 1 & 0 & 1 & 10 & 1 & 1 & 10 \\ & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline\hline & & 1 & 1 & 1 & 0 & 1 \end{array}$$

Step 4 We replace the row representing the number we are subtracting *from* by the row above the horizontal line and complete the subtraction.

$$\begin{array}{cccccc} 10 & 1 & 10 & 1 & 1 & 10 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ \hline\hline 1 & 1 & 1 & 1 & 0 & 1 \end{array}$$

Thus we have obtained $(1101000)^2 - (101011)^2 = (111101)_2$. In decimal numbers, this subtraction is $104 - 43$ and the answer we obtained is 61. □

Binary multiplication

The process of binary multiplication is very similar to long multiplication in the decimal system. A familiar feature in decimal is that when we multiply any whole number by 10, we move all the digits one place to the left and enter a zero in the units column. Let us pause a moment to see why this rule works.

Example 1.8 Consider 743×10. Writing 743 in expanded notation, we have

$$743 = 7(10^2) + 4(10^1) + 3(1).$$

Multiplying each term in this equation by 10, we obtain

$$\begin{aligned} 743 \times 10 &= 7(10 \times 10^2) + 4(10 \times 10^1) + 3(10 \times 1) \\ &= 7(10^3) + 4(10^2) + 3(10) \\ &= 7(10^3) + 4(10^2) + 3(10^1) + 0(1). \end{aligned}$$

Thus the *place value* of each digit has been multiplied by 10 and the the entry in the units column is now 0. □

A very similar thing happens when we multiply a binary number by the base number $2 = (10)_2$. As we have seen in Example 1.4, the binary number $(a_4a_3a_2a_1a_0)_2$ can be written in expanded notation as

$$a_4(2^4) + a_3(2^3) + a_2(2^2) + a_1(2^1) + a_0(1).$$

Multiplying this number by 2, we obtain

$$\begin{aligned} & a_4(2 \times 2^4) + a_3(2 \times 2^3) + a_2(2 \times 2^2) + a_1(2 \times 2^1) + a_0(2 \times 1) \\ = \; & a_4(2^5) + a_3(2^4) + a_2(2^3) + a_1(2^2) + a_0(2) \\ = \; & a_4(2^5) + a_3(2^4) + a_2(2^3) + a_1(2^2) + a_0(2^1) + 0(1). \end{aligned}$$

Thus the place value of each bit has been multiplied by 2 and so the bits all move one place to the left and the entry in the units column is 0. Thus we have the following rule:

Rule 1.3 *To multiply by $(10)_2$ in binary, we move each bit one place to the left and enter 0 in the 2^0 column.*

Example 1.9 Multiplying $(1101)_2 \times (10)_2$, the rule gives $(11010)_2$. Checking in decimal, we have calculated 13×2 and obtained 26. □

We can extend the rule for multiplying by 2 to multiplying by any power of 2. For example, since multiplying by 4 is the same as multiplying by 2 and then multiplying the result by 2 again, we move each bit *two* places to the left and enter zeros in both the 2^1 and the 2^0 columns. Thus the rule for multiplying by $4 = (100)_2$ in binary is exactly the same as the rule for multiplying by 100 in decimal. Similarly, the rule for multiplying by $8 = (1000)_2$ in binary is the same as the rule for multiplying by 1000 in decimal, and so on.

Example 1.10 We multiply $(1011)_2 \times (101)_2$. Note first that $(101)_2 = (100)_2 + (1)_2$. So we

perform two multiplications and add them together.

		1	0	1	1	
			1	0	1	
1	0	1	1	0	0	$(1011)_2 \times (100)_2$
		1	0	1	1	$(1011)_2 \times (1)_2$
	1					*bits carried forward in the sum*
1	1	0	1	1	1	$(1011)_2 \times (101)_2$

Thus we have $(1011)_2 \times (101)_2 = (110111)_2$. We leave you to check this is correct by finding the decimal equivalent. □

Binary division

We first note that dividing by 2 is just the reverse of the process of multiplying by 2 and so we have the following rule.

Rule 1.4 *To divide by $(10)_2$ in binary, we move each bit one place to the right and the entry in the units column becomes the remainder.*

Example 1.11 $(110111)_2 \div (10)_2 = (11011)_2$, remainder $(1)_2$. □

The number we are dividing by is called the **divisor** and the number of complete times it divides is called the **quotient**. In the example above, the divisor is $(10)_2$ and the quotient is $(11011)_2$.

We can easily extend Rule 1.4 to powers of 2, as illustrated in the following example.

Example 1.12 $(110111)_2 \div (1000)_2 = (110)_2$, remainder $(111)_2$. □

Note that to retrieve the number we are dividing *into*, we reverse the process by multiplying the quotient by the divisor and adding the remainder. Thus the reverse of the process in Example 1.11 is

$$(110111)_2 = (11011)_2 \times (10)_2 + (1)_2.$$

The process of long division in the binary system is very similar to long division in the decimal system, but simplified by the fact that when we divide a binary number by a divisor with the *same* number of bits, the quotient is either $(1)_2$ or $(0)_2$. In the latter case, when we bring down the next bit, we will always obtain a quotient of $(1)_2$. The process is illustrated in the following example.

Example 1.13 We calculate $(101001)_2 \div (11)_2$.

Step 1 We try dividing $(11)_2$ into the first two bits. Since $(11)_2$ is greater than $(10)_2$, the quotient is 0_2 and we bring down the next bit, so that we are now dividing $(11)_2$ into $(101)_2$. The quotient this time is $(1)_2$, which we record in the quotient line above the third bit. We now subtract $(11)_2 \times (1)_2 = (11)_2$ from the first three bits to obtain the remainder $(10)_2$.

			1			
11	1	0	1	0	0	1
		1	1			
		1	0			

Step 2 We bring down the next bit. We are now dividing a 2-bit number into a 3-bit number, so the quotient must be $(1)_2$. We repeat the process described in Step 1.

			1	1		
11	1	0	1	0	0	1
		1	1			
		1	0	0		
			1	1		
				1		

Step 3 We bring down the next bit. Then since $(11)_2$ is greater than $(10)_2$, we record (0_2) in the quotient row and bring down the last bit. The quotient must be $(1)_2$ this time, so we record $(1)_2$ above the last bit in the quotient row and subtract $(11)_2$ from $(101)_2$ to give the remainder.

$$
\begin{array}{r|cccccc}
 & & & 1 & 1 & 0 & 1 \\
\hline
11 & 1 & 0 & 1 & 0 & 0 & 1 \\
 & & 1 & 1 & & & \\
\cline{2-4}
 & & 1 & 0 & 0 & & \\
 & & & 1 & 1 & & \\
\cline{3-5}
 & & & & 1 & 0 & 1 \\
 & & & & & 1 & 1 \\
\cline{4-6}
 & & & & & 1 & 0
\end{array}
$$

Thus we have obtained $(101001)_2 \div (11)_2 = (1101)_2$ with remainder $(10)_2$. Checking in decimal, we have divided 41 by 3 to obtain 13 with remainder 2. □

1.1.4 The hexadecimal number system

Another number system commonly used in computer science is the **hexadecimal** system based on 16. This system has the following advantages: it enables most numbers to be recorded using substantially fewer digits than are necessary in base 2; there is a very easy method of converting between hexadecimal and binary, which as we have seen is a "natural" language for computers to use.

In order to express a number in the hexadecimal system we need 16 digits. It is customary to use the digits $0, 1, \ldots, 9$ followed by the letters A,B,C,D,E, and F as symbols to represent the numbers $10, 11, 12, 13, 14$ and 15 respectively. The hexadecimal digits are known as **hexits** and the hexadecimal system is often called **hex**, for short.

We convert from hex to decimal in the usual way, using expanded notation in hexits.

Example 1.14 We first illustrate the number $(5AE7)_{16}$ using hexadecimal boxes.

16^3	16^2	16^1	1
5	A	E	7

Writing it in expanded notation, we have

$$
\begin{aligned}
(5AE7)_{16} &= 5(16^3) + 10(16^2) + 14(16^1) + 7(1) \\
&= 23271. \ \square
\end{aligned}
$$

Arithmetic in hex

Hexadecimal arithmetic is very similar to decimal arithmetic, except that we have 16 hexits and carry, or borrow, 16's instead of 10's. We work two illustrative examples here.

Example 1.15 We add $(9D7)_{16} + (4BA)_{16}$. Hexits carried into the next column are entered in the row above the horizontal line.

$$
\begin{array}{ccc}
1 & 1 & \\
\hline
9 & D & 7 \\
4 & B & A \\
\hline\hline
E & 9 & 1
\end{array}
$$

Thus $(9D7)_{16} + (4BA)_{16} = (E91)_{16}$. The decimal equivalent is $2519 + 1210 = 3729$. □

Example 1.16 We subtract $(F04)_{16} - (CD9)_{16}$. The revised row when we have borrowed a 16 is shown in the row above the horizontal line. Note that all the numbers in the table are in hex, so that the "14" in the revised row is $(14)_{16}$, representing the decimal number 20.

$$
\begin{array}{ccc}
E & F & 14 \\
\hline
F & 0 & 4 \\
C & D & 9 \\
\hline\hline
2 & 2 & B
\end{array}
$$

Thus $(F04)_{16} - (CD9)_{16} = (22B)_{16}$. The decimal equivalent is $3844 - 3289 = 555$. □

Converting between hex and binary

There is a very easy procedure for converting between the binary number and hexadecimal number systems. First, we need a table giving the binary equivalents of each hexit. You will see that we have expressed each binary number as a 4-bit string, by adding zeros where necessary at the *front* of each binary number to make up the total number of bits to four.

hexadecimal	$(0)_{16}$	$(1)_{16}$	$(2)_{16}$	$(3)_{16}$	$(4)_{16}$	$(5)_{16}$	$(6)_{16}$	$(7)_{16}$
4-bit binary	$(0000)_2$	$(0001)_2$	$(0010)_2$	$(0011)_2$	$(0100)_2$	$(0101)_2$	$(0110)_2$	$(0111)_2$

hexadecimal	$(8)_{16}$	$(9)_{16}$	$(A)_{16}$	$(B)_{16}$	$(C)_{16}$	$(D)_{16}$	$(E)_{16}$	$(F)_{16}$
4-bit binary	$(1000)_2$	$(1001)_2$	$(1010)_2$	$(1011)_2$	$(1100)_2$	$(1101)_2$	$(1110)_2$	$(1111)_2$

To convert a number in binary into hex, we first split the binary string into blocks of four bits, *starting from the bit in the units column and working to the left.* If the total number of bits is not divisible by 4, then we add the necessary number of zeros to the *front* of the number to complete the last block on the left. Each block of four bits now corresponds to a hexit, as shown in the table above. We simply replace each 4-bit block by the corresponding hexit. To convert from hex to binary, we reverse this process and replace each hexit by a 4-bit binary string representing the corresponding binary number.

Example 1.17 We write the binary number $(1101101011)_2$ in hex. Splitting the binary string into blocks of four bits, working from the units column towards the left, we have

$$1101101011 = 11\ \ 0110\ \ 1011.$$

The first block is incomplete, so we add two zeros on the front to complete it. This gives

$$1101101011 = 0011\ \ 0110\ \ 1011.$$

Finally, replacing each block of four bits by the corresponding hexit, we have

$$(1101101011)_2 = (36B)_{16}. \ \square$$

Example 1.18 We express $(B35)_{16}$ in binary. Replacing each hexit by the corresponding 4-bit binary string, we obtain

$$1011\ \ 0011\ \ 0101$$

Thus we have

$$(B35)_{16} = (101100110101)_2. \ \square$$

1.1.5 Converting decimal integers into other bases

Consider the binary number $(a_4a_3a_2a_1a_0)_2$, where each of the symbols $a_0, a_1, \ldots$ represents a bit (0 or 1). By Rule 1.4, when we divide a binary number by $2 = (10)_2$, all the bits move one place to the right and the bit in the units column becomes the remainder. Thus

$$(a_4a_3a_2a_1a_0)_2 \div (10)_2 = (a_4a_3a_2a_1)_2, \textit{ remainder } a_0.$$

So, to write a decimal number in binary, we start by dividing the decimal number by 2 and the remainder, 0 or 1, gives the value of the units bit a_0.

We now repeat this process with the quotient in place of the original number to find the value of the bit a_1. Thus

$$(a_4a_3a_2a_1)_2 \div (10)_2 = (a_4a_3a_2)_2, \textit{ remainder } a_1,$$

so that the remainder at this second division by 2 gives the value of a_1. We continue this process until we have found the value of each bit.

Example 1.19 We express the decimal integer 25 in binary.

$$\begin{aligned} 25 \div 2 &= 12, \quad \textit{remainder } 1 \\ 12 \div 2 &= 6, \quad \textit{remainder } 0 \\ 6 \div 2 &= 3, \quad \textit{remainder } 0 \\ 3 \div 2 &= 1, \quad \textit{remainder } 1 \\ 1 \div 2 &= 0, \quad \textit{remainder } 1. \end{aligned}$$

Now the first remainder gives us the value of the bit in the units column, the second remainder gives us the value of the bit in the 2^1 column, the next remainder gives the bit in the 2^2 column, and so on. Thus, to find the binary representation of 25, we must read off the remainders in REVERSE ORDER. This gives

$$25 = (11001)_2. \square$$

We can use this method to convert a decimal number to any base.

Example 1.20 We convert 471 to base 5.

$$\begin{aligned} 471 \div 5 &= 94, \quad \textit{remainder } 1 \\ 94 \div 5 &= 18, \quad \textit{remainder } 4 \\ 18 \div 5 &= 3, \quad \textit{remainder } 3 \\ 3 \div 5 &= 0, \quad \textit{remainder } 3 \end{aligned}$$

Recording the remainders in reverse order, gives

$$471 = (3341)_5. \square$$

Example 1.21 We convert 2963 to hexadecimal.

$$\begin{aligned} 2963 \div 16 &= 185, \quad \textit{remainder } 3 \\ 185 \div 16 &= 11, \quad \textit{remainder } 9 \\ 11 \div 16 &= 0, \quad \textit{remainder } 11 \end{aligned}$$

Recording the remainders in reverse order, gives

$$2963 = (B93)_{16}. \square$$

1.2 Rational numbers

Learning Objectives

After studying this section, you should be able to:

- factorize a given decimal integer into its prime factors and use power notation;
- say what is meant by a *rational number* and express a recurring decimal as a fraction;
- interpret fractions in bases other than 10.

Introduction

Thus far, we have considered only the representation of positive integers in various number bases. In this section, we consider division of positive integers and the representation of fractions.

1.2.1 Factors, multiples and primes

Definition 1.5 *Suppose that m and n are positive integers. If n divides into m exactly, so that the quotient m/n is also an integer, then we say that* **n divides m**. *We also say that* **n is a factor** *(or a* **divisor***)* **of m**, *and that* **m is a multiple of n**.

Thus, for example, we can say that "4 divides 20", "4 is a factor of 20", "20 is a multiple of 4". These three sentences all have the same meaning.

Definition 1.6 *A* **prime number** *(or more simply, a* **prime***) is an integer greater than 1 whose only positive factors are itself and 1.*

Notice carefully that the number 1 is *not* a prime. It is called a **unit** and has the property that it divides *every* integer. A number greater than 1 that is not prime is called **composite**. The smallest primes are thus $2, 3, 5, 7, 11, 13, 17, \ldots$, whereas the numbers $4, 6, 8, 9, 10, 12, 14, 15, 16, \ldots$ are composite.

Example 1.22 The composite number 30 can be written as a product of two positive integers greater than 1 in several different ways. For example, we could write

$$30 = 2 \times 15,\ 30 = 3 \times 10, \text{ or } 30 = 5 \times 6.$$

Suppose we now repeat the process and express the composite integers in these products as products of factors, until every factor in each product is a prime. We would then obtain

$$\begin{aligned} 2 \times 15 &= 2 \times 3 \times 5 \\ 3 \times 10 &= 3 \times 5 \times 2 \\ 5 \times 6 &= 5 \times 2 \times 3. \end{aligned}$$

Note that each way of splitting up 30 gives the *same* set of prime factors; the final expressions differ only in the order which the primes are listed. □

Example 1.22 illustrates the following general result. You should be familiar with its statement, but the proof is beyond the scope of this course.

Theorem 1.7 The Fundamental Theorem of Arithmetic *Every positive integer greater than 1 can be written as the product of 1 and prime factors. Further, this expression is unique, except for the order in which the prime factors occur.*

The only reason for the appearance of 1 in this theorem is to accomodate the case when the integer is itself a prime number, such as 29 for example. It does not make sense to say that 29 is a product of 29, but we can say that 29 is a product 1 and 29. When asked to express a composite integer as a product of prime factors, we omit the unit 1. This theorem guarantees that however we go about splitting up an integer into prime factors, we will always obtain the same prime factors.

Example 1.23 We write 280 as the product of its prime factors. From Theorem 1.7, we can start by writing 280 as the product of any two of its factors. An obvious factor of 280 is 10, and so we can proceed as follows:

$$\begin{aligned} 280 &= 10 \times 28 \\ &= (2 \times 5) \times (4 \times 7) \\ &= 2 \times 5 \times (2 \times 2) \times 7. \end{aligned}$$

To simplify the expression, we gather together all occurences of the same prime and write the product as

$$280 = 2^3 \times 5 \times 7,$$

using index notation to deal with repeated factors, and listing the distinct primes in ascending order of size. □

Since every positive integer corresponds in a unique way to its prime factors, the primes can be regarded as the basic building blocks of the integers. The study of primes and the properties of

the integers has a long history, going back at least 2500 years to the early Greek mathematicians of the School of Pythagoras. An important modern application of prime numbers is to Public Key Cryptography. The details of this application are outside the scope of this course[1].

1.2.2 Representing fractions

Definition 1.8 *Numbers that can be expressed as a fraction (or ratio) m/n of two integers m and n, where $n \neq 0$, are called* **rational** *numbers.*

We have two ways of representing a rational number, whatever the number base. One is to write it in the form of a fraction m/n, where both m and n are integers (expressed in the relevant number base) and $n \neq 0$. Note that all integers are special cases of rational numbers, since we can write an integer m as the fraction $m/1$.

The other method of representing a rational number is to write it in the form of a decimal, using *negative* as well as positive and zero powers of the base as place values. We shall see in this section that the decimal form of a rational number has a very special property.

1.2.3 Decimal fractions

Let us look first at the representation of rational numbers in base 10 in expanded notation. The place values are illustrated by the following boxes.

$\cdots$	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	$\cdots$

You will be familiar with the process of turning a fraction m/n into a decimal by dividing n into m. Since we are interested in what happens to the part of the expansion after the decimal point, we shall assume that m and n are positive integers with m less than n.

The decimal expansion of a fraction has an interesting feature: it either terminates after a finite number of places, as for example in the expansion $17/20 = 0.85$ or $3/8 = 0.375$; or *it has a recurring block of figures*, as for example the digit 6 in the expansion $5/12 = 0.41666...$ or the block 63 in $7/11 = 0.636363....$ We investigate why this occurs.

It can be shown that when we divide by a positive integer n, we can always choose the quotient so that the remainder is either 0 or one of the positive integers less than n. There are thus only n possible remainders when we divide by n. Now when we divide $m.000\cdots$ by n, the remainder at each division, becomes the "carrying number" for the next division. Since we can continue the divisions indefinitely and *there are only n possible carrying numbers*, we either obtain (eventually) a remainder 0, in which case the decimal expansion terminates, or *we eventually obtain a remainder that has occurred before*. From then on, the same sequence of quotients and remainders will occur again and again.

Example 1.24 Consider the decimal expansion of $4/7$. On dividing by 7, the remainder is always one of the six possibilities: $0, 1, 2, 3, 4, 5$ or 6.

$$\begin{array}{r|ccccccccc} & 0. & 5 & 7 & 1 & 4 & 2 & 8 & 5 & 7 & \cdots \\ \hline 7 & 4. & {}^{4}0 & {}^{5}0 & {}^{1}0 & {}^{3}0 & {}^{2}0 & {}^{6}0 & {}^{4}0 & {}^{5}0 & \cdots \end{array}$$

As you see, in this case, all six possible remainders arise: the first is 4, then we have successively remainders of $5, 1, 3, 2, 6$; then the remainder 4 occurs again. From this point on, the same sequence of six remainders and quotients repeats itself. Thus the decimal expansion of $4/7$ has a repeated block of six digits. □

We now consider the converse problem. Suppose we are given a terminating or repeating decimal. Is it possible to write it as a fraction m/n, where m and n are integers? Clearly this is possible in the case of a terminating decimal, since we can write it with a denominator that is a power of 10. By a neat trick, we can show that any recurring decimal can also be expressed as a fraction m/n and find the values of m and n. We explain the method in the context of the following examples.

[1] If you are interested in reading about this application, there is a clear and interesting explanation in *Discrete Mathematics with Applications*, by M.O. Albertson and J.P. Hutchings (see Appendix A).

Example 1.25 We express the repeating decimals 0.272727... as a fraction in its lowest terms.

Step 1 Determine the number of digits in the repeating block. In this case, the repeating block is 27 and so the number of digits is 2.

Step 2 Write $x = 0.272727...$ and multiply x by 10^2, where the power of 10 is the length of the repeating block. This gives

$$100x = 27.272727\ldots.$$

Step 3 Subtract x from $100x$.

$$\begin{array}{rrcl} & 100x & = & 27.272727\ldots \\ - & x & = & 0.272727\ldots \\ \hline & 99x & = & 27.000000\ldots \end{array}$$

Step 4 Solve the equation $99x = 27$ to find x. This gives $x = 27/99$. Cancelling the common factor of 9, we have $x = 3/11$ as a fraction in its lowest terms. □

Example 1.26 We express the repeating decimal 0.15333... as a fraction in its lowest terms.

Step 1 Determine the number of digits in the repeating block. In this case, the repeating block is the single digit 3 and so the number of digits is 1.

Step 2 Write $x = 0.15333...$ and multiply x by 10, where the power of 10 is the length of the repeating block. This gives

$$10x = 1.53333\ldots.$$

Step 3 Subtract x from $10x$.

$$\begin{array}{rrcl} & 10x & = & 1.53333\ldots \\ - & x & = & 0.15333\ldots \\ \hline & 9x & = & 1.38000\ldots \end{array}$$

Step 4 Solve the equation $9x = 1.38$. We obtain $x = 1.38/9 = 138/900 = 23/150$. □

Since this process can be performed on any decimal with a recurring block, we see that any recurring decimal represents a fraction of two integers. Thus we have the following important result.

Result 1.9 *The rational numbers are just those numbers that can be represented as a terminating or recurring decimal.*

1.2.4 Fractions in bases other than 10

Binary system

We can represent rational numbers in any base by using negative powers of the base as place values. The place values in the full binary system are illustrated below.

$\cdots$	2^3	2^2	2^1	$2^0 = 1$	2^{-1}	2^{-2}	2^{-3}	$\cdots$

Example 1.27 The binary number $(0.1101)_2$ can be expressed in base 10 as follows.

$$1(2^{-1}) + 1(2^{-2}) + 0(2^{-3}) + 1(2^{-4}) = 1/2 + 1/4 + 1/16 = 13/16 = 0.8125.$$

An alternative method, is first to express $(0.1101)_2$ as a binary fraction, and then to convert both the numerator and denominator to base 10. Thus we have

$$(0.1101)_2 = (1101)/(10000)_2 = 13/16 = 0.8125.$$

The binary number $(101.11)_2$ represents

$$1(2^2) + 0(2^1) + 1(1) + 1(2^{-1}) + 1(2^{-2}) = 2^2 + 2^0 + 2^{-1} + 2^{-2} = 5.75.$$

Alternatively, we could work as follows.

$$(101.11)_2 = (101)_2 + (11/100)_2 = 5 + 3/4 = 5.75. \square$$

The digits to the *left* of the point are called the **integral part** of the number and the digits to the *right* of the point are called the **fractional part** of the number. Thus, for example, the integral part of $(101.11)_2$ is $(101)_2$ and the fractional part is $(0.11)_2$.

Hexadecimal system

The place values in the full hexadecimal system are illustrated below.

$\cdots$	16^3	16^2	16^1	$16^0 = 1$	16^{-1}	16^{-2}	16^{-3}	$\cdots$

Example 1.28 The number $(E7.3B)_{16}$ represents the decimal number

$$14(16^1) + 7(1) + 3(16^{-1}) + 11(16^{-2}) = 231 + 3/16 + 11/256 = 231.23046875. \square$$

To convert the fractional part of a binary number into the hexadecimal system, we split the binary string into groups of four bits *starting from the decimal point and working to the right.* If the last block to the right contains less than four bits, we add zeros as necessary *on the end of the number* to complete the block. We then use the binary-hex conversion table given in section 1.1.4.

Example 1.29 Converting the binary number $(101011101.11)_2$ to hex, we have

$$101011101.11 = 0001\ \ 0101\ \ 1101\ \ .1100 = (15D.C)_{16}.$$

Converting the hex number $(B3.5)_{16}$ to binary, we have

$$B3.5 = 1011\ 0011.0101 = (10110011.0101)_2. \square$$

1.3 Real numbers

Learning Objectives

After studying this section, you should be able to:

- give examples of the kinds of measurements for which the real number system is used;
- give examples of irrational numbers;
- use and interpret *greater than* and *less than* signs;
- express a rational number in floating point form and scientific notation.

Introduction

It is not easy to define exactly what we mean by a **real** number. We can visualize them geometically as a set of numbers that can be put into one-to-one correspondence with the points of a line, infinite in extent. This is the procedure we adopt when we mark a scale on a coordinate axis in order to graph a function. The real numbers have the important property that they form what is called a *continuum*. The set of real numbers is therefore used when we want to measure a quantity such as time, length or speed that can vary continuously.

1.3.1 Irrational numbers

The question arises as to whether all real numbers are rational, or whether there exist numbers that cannot be expressed as a fraction of two integers. The ancient Greek mathematicians answered this question in the 4th Century BC by showing that the square root of any integer that is not an exact square cannot be expressed as a fraction. This shows the existence of real numbers that are not rational. We call such numbers **irrational**. As well as numbers like $\sqrt{2}$, $\sqrt[3]{5}$, etc, numbers such as π and e (base of the natural logarithm) are irrational numbers. It follows from Result 1.9, that every irrational number has a decimal expansion that neither repeats nor terminates. Thus we can only ever give an approximation to the value of an irrational number in the decimal number system, or indeed in any number system with an integer base.

1.3.2 Inequality symbols

Suppose x and y are two different real numbers. The fact that x is not equal to y is expressed symbollically as

$$x \neq y.$$

Note that the word *distinct* is often used in mathematics, rather than "different", to describe two or more objects of the same class that can be distinguished from one another. So, in this case, we could have described x and y as "distinct numbers".

If $x \neq y$, then one of x and y must be the greater of the two, say x is greater than y. We write this as

$$x > y.$$

An equivalent statement is "y is less than x", written

$$y < x.$$

To avoid confusing these symbols, note that the pointed end (or small end) of either inequality sign always points towards the smaller number, while the open (or larger) end points to the larger number.

Example 1.30 The decimal expansion of π begins $3.1415\ldots$. Thus we can say that $\pi > 3.14$ and also that $\pi < 3.15$. Together, these two inequalities tell us that the true value of π is between 3.14 and 3.15. We can express this by concatenating the two inequalities as follows:

$$3.14 < \pi < 3.15.$$

Notice that we have turned the inequality $\pi > 3.14$ round and expressed it as $3.14 < \pi$, before combining it with the other inequality, so that π is sandwiched between the smallest and the largest of the three numbers and both inequality signs point in the same direction. □

Example 1.31 The decimal expansion of 17/9 is $1.888\ldots$. Thus 17/9 lies between 1.888 and 1.889. We can express this symbolically by writing

$$1.888 < 17/9 < 1.889. \ \square$$

The symbol ">" can be modified to express *at least* or, equivalently, *greater than or equal to*. For example,

$$x \geq 5.3$$

is read "x is at least 5.3", or "x is greater than or equal to 5.3". Similarly, the symbol $\leq$' reads "at most" or "less than or equal to".

Example 1.32 Suppose a real number x satisfies the inequalities $14 \leq x \leq 25$. The inequality $14 \leq x$ tells us that x is *at least* 14. The inequality $x \leq 25$ means that x is *at most* 25. Thus together they read: "x is at least 14 and at most 25".

The inequality sign "$\leq$" can also be concatenated with "<", in either order. So, for example, $14 \leq x < 25$ means that x is at least 14 and less than 25. □

1.3.3 Floating-point notation

A digital device, such as a hand-held calculator or digital computer, is only able to store a finite (and often fixed) number of digits to represent any numeral. On the other hand, all irrational numbers and rational numbers with repeating blocks in their decimal expansion require an infinite number of digits to represent them exactly in decimal form. Even a terminating decimal may have more digits in its decimal expansion than the device is able to store. Thus most real numbers will be represented in a digital computer only by an approximation, formed by cutting off the decimal expansion after a certain number of digits. This produces an error in calculations known as a **truncation error**.

Suppose that a calculator can store only 10 digits. Then, for example, the error in representing 5/12 as 0.416666666 is 0.000000000666... ., or a little less than 0.000000000667. This may not seem very grave, but in a complex calculation errors may accumulate, giving a substantial error in the solution.

It is hard to read and comprehend a number such as 0.000000000667 written in this form. There are more convenient ways of expressing very small or very large decimal numbers. The two main ones are called *scientific notation* and *floating-point notation*. Most computers can store real numbers in **floating-point notation**, so we shall describe this method in detail. In this system, a positive number is expressed as the product of a power of 10 called the **exponent**, and a number x such that $0.1 \leq x < 1$ called the **mantissa**.

Example 1.33 In floating point notation, the number 0.000000000667 is expressed as 0.667×10^{-9}, where -9 is the exponent and 0.667 is the mantissa; the number 146.75 is expressed as 0.14675×10^3, where 3 is the exponent and 0.14675 is the mantissa. □

Scientific notation is based on a similar principle, and differs only in that the mantissa is a number x such that $1 \leq x < 10$.

Example 1.34 In scientific notation, the number 0.000000000667 is expressed as 6.67×10^{-10}, where -10 is the exponent and 6.67 is the mantissa; the number 146.75 is expressed as 1.4675×10^2, where 2 is the exponent and 1.4675 is the mantissa. □

1.4 Exercises 1

1. (a) Write the decimal numbers 4037 and 40371 in expanded form as multiples of powers of 10.

   (b) Suppose that a is the 4-digit decimal number $a_3a_2a_1a_0$. Write a in expanded form as multiples of powers of 10. Suppose that x is the 5-digit decimal number $a_3a_2a_1a_01$. Can you express x in terms of a?

2. (a) A number is expressed in base 5 as $(234)_5$. What is it as a decimal number?

   (b) Suppose you multiply $(234)_5$ by 5. What would the answer be in base 5?

3. Express the following binary numbers as decimal numbers:

$$(11011)_2; \qquad (1100110)_2; \qquad (111111111)_2.$$

   Can you think of a quick way of doing the last one?

4. Perform the binary additions

$$(10111)_2 + (111010)_2; \qquad (1101)_2 + (1011)_2 + (1111)_2.$$

5. Perform the binary subtractions

$$(1001)_2 - (111)_2; \qquad (110000)_2 - (10111)_2.$$

6. Perform the following binary multiplications

$$(1101)_2 \times (101)_2; \qquad (1101)_2 \times (1101)_2.$$

7. Perform the binary division $(111011)_2 \div (101)_2$, giving a quotient and remainder.

8. Find the decimal equivalents of each of the binary numbers in the previous questions and check your answers by decimal arithemetic.

9. (a) Write the hexadecimal numeral $(A5D)_{16}$ in decimal.
   (b) Suppose that a is represented in hex by $(a_2a_1a_0)_{16}$. Write a in expanded form as multiples of powers of 16. Suppose that x and y are represented in hex by $(a_2a_1a_00)_{16}$ and $(a_2a_1a_0A)_{16}$, respectively. Can you express x and y in terms of a in base 10?

10. Calculate $(BBB)_{16} + (A5D)_{16}$ and $(BBB)_{16} - (A5D)_{16}$, working in hex.

11. (a) Write the hex number $(EC4)_{16}$ in binary.
    (b) Write the binary number $(1111011010)_2$ in hex.

12. Express the decimal number 753 (a) in binary; (b) in base 5; (c) in hex.

13. Express 42900 as a product of its prime factors, using index notation for repeated factors.

14. Express the recurring decimals 0.126126126... and 0.7545454... as fractions *in their lowest terms.*

15. Given that π is an irrational number, can you say whether $\frac{\pi}{2}$ is rational or irrational? Or is it impossible to tell?

16. (a) Express the binary number $(0.101)_2$ in base 10 and in base 16.
    (b) Express the hex number $(B.25)_{16}$ in base 10 and in base 2.
    (c) Write the decimal fraction 3/8 in base 2 (as a "bicimal").
    (d) Can you work out how to express the number $(0.32)_5$ in base 10?

17. Express the following inequalities in symbols:
    (a) 5/7 lies between 0.714 and 0.715;
    (b) $\sqrt{2}$ is at least 1.41;
    (c) $\sqrt{3}$ is at least 1.732 and at most 1.7322.

18. Write the numbers 0.0000526 and 429000000 in floating point form. How is the number 1 expressed in this notation?

19. Let $n = ab$ be a composite integer, where a, b are proper factors of n and $a \leq b$.
    (a) Say why $a \leq \sqrt{n}$.
    (b) Deduce that every composite integer n has a prime factor p such that $p \leq \sqrt{n}$.
    (c) Use this result to prove that 89 is prime by testing it for divisibility by just four prime numbers.
    (d) Decide whether 899 is prime.

20. (a) What is the *maximum* number of digits that a decimal fraction with denominator 13 could have in a recurring block in theory?
    (b) How many digits does 1/13 actually have?
    (c) Do each of the decimal fractions $1/13, 2/13, 3/13, 4/13, 5/13$ have the same number of digits in a recurring block? What do you notice about the digits in the recurring blocks of the fractions $1/13, 3/13, 4/13$?
    (d) Can you predict which other fractions with denominator 13 will have the same digits as 1/13 in their recurring block?

Appendix B

Solutions to Exercises

Exercises 1

1. (a) $4(10^3)+3(10)+7$; $4(10^4)+3(10^2)+7(10)+1(1)$; (b) $a = a_3(10^3)+a_2(10^2)+a_1(10)+a_0(1)$; $x = a_3(10^4) + a_2(10^3) + a_1(10^2) + a_0(10) + 1$; $x = 10a + 1$.

2. (a) $2(5^2) + 3(5) + 4 = 69$; (b) $(2340)_5$.

3. 27; 102; 511. The quick way is to see that $(111111111)_2 = 2^9 - 1$.

4.

$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 1 & & \\ \hline & & 1 & 0 & 1 & 1 & 1 \\ & 1 & 1 & 1 & 0 & 1 & 0 \\ \hline\hline 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \qquad \begin{array}{cccccc} 1 & 0 & 1 & 1 & 1 & \\ \hline & & 1 & 1 & 0 & 1 \\ & & 1 & 0 & 1 & 1 \\ & & 1 & 1 & 1 & 1 \\ \hline\hline 1 & 0 & 0 & 1 & 1 & 1 \end{array}$$

Thus (a) $(10111)_2 + (111010)_2 = (1010001)_2$; (b) $(1101)_2 + (1011)_2 + (1111)_2 = (100111)_2$.

5.

$$\begin{array}{cccc} 0 & 1 & 10 & \\ \hline 1 & 0 & 0 & 1 \\ & 1 & 1 & 1 \\ \hline\hline 0 & 0 & 1 & 0 \end{array} \qquad \begin{array}{cccccc} 0 & 10 & 1 & 1 & 1 & 10 \\ \hline 1 & 1 & 0 & 0 & 0 & 0 \\ & 1 & 0 & 1 & 1 & 1 \\ \hline\hline 0 & 1 & 1 & 0 & 0 & 1 \end{array}$$

Hence (a) $(1001)_2 - (111)_2 = (10)_2$; (b) $(110000)_2 - (10111)_2 = (11001)_2$.

6.

$$\begin{array}{ccccccc} & & & 1 & 1 & 0 & 1 \\ & & & & 1 & 0 & 1 \\ \hline & 1 & 1 & 0 & 1 & 0 & 0 \\ & & & 1 & 1 & 0 & 1 \\ \hline\hline 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \qquad \begin{array}{cccccccc} & & & & 1 & 1 & 0 & 1 \\ & & & & 1 & 1 & 0 & 1 \\ \hline & & 1 & 1 & 0 & 1 & 0 & 0 \\ & & & 1 & 1 & 0 & 1 & 0 \\ & & & & 1 & 1 & 0 & 1 \\ \hline\hline 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{array}$$

Hence (a) $(1101)_2 \times (101)_2 = (1000001)_2$; (b) $(1101)_2 \times (1101)_2 = (10101001)_2$.

7.

$$\begin{array}{r|cccccc} & & & 1 & 0 & 1 & 1 \\ \hline 101 & 1 & 1 & 1 & 0 & 1 & 1 \\ & 1 & 0 & 1 & & & \\ \cline{2-4} & & 1 & 0 & 0 & 1 & \\ & & & 1 & 0 & 1 & \\ \cline{3-5} & & & 1 & 0 & 0 & 1 \\ & & & & 1 & 0 & 1 \\ \cline{4-6} & & & & 1 & 0 & 0 \end{array}$$

Hence $(111011)_2 \div (101)_2 = (1011)_2$, remainder $(100)_2$.

8. $23 + 58 = 81$, $13 + 11 + 15 = 39$; $9 - 7 = 2$, $48 - 23 = 25$; $13 \times 5 = 65$, $13 \times 13 = 169$; $59 \div 5 = 11$, remainder 4.

9. (a) 2653; (b) $a = a_2(16^2) + a_1(16) + a_0(1)$; $x = 16a$, $y = 16a + 10$.

10.

$$\begin{array}{cccc} 1 & 1 & 1 & \\ \hline & B & B & B \\ & A & 5 & D \\ \hline\hline 1 & 6 & 1 & 8 \end{array} \qquad \begin{array}{ccc} & A & 1B \\ \hline B & B & B \\ A & 5 & D \\ \hline\hline 1 & 5 & E \end{array}$$

Hence $(\mathrm{BBB})_{16} + (\mathrm{A5D})_{16} = (1618)_{16}$; $(\mathrm{BBB})_{16} - (\mathrm{A5D})_{16} = (15\mathrm{E})_{16}$.

11. (a) $(111011000100)_2$; (b) $(3\mathrm{DA})_{16}$.

12. (a) $(1011110001)_2$; (b) $(11003)_5$; (c) $(2\mathrm{F1})_{16}$.

13. $42900 = 2^2 \times 3 \times 5^2 \times 11 \times 13$.

14. (a) Let $x = 0.126126126\ldots$. Then $1000x = 126.126126126\ldots$. Subtracting x gives $999x = 126$, so that $x = 126/999 = 14/111$.

(b) Let $x = 0.7545454\ldots$. Then $100x = 75.454545\ldots$. Subtracting x gives $99x = 74.7$. Hence $x = 74.7/99 = 747/990 = 83/110$.

15. $\frac{\pi}{2}$ is irrational. For, suppose $\frac{\pi}{2}$ is rational. Then we can write $\frac{\pi}{2} = \frac{a}{b}$, where a, b are integers. But this gives $\pi = \frac{2a}{b}$, which implies π is rational also, which we know is false.

16. (a) $5/8 = 0.625$; $(0.\mathrm{A})_{16}$; (b) $11 + \frac{2}{16} + \frac{5}{256} = 11.14453125$; $(1011.00100101)_2$; (c) $(0.011)_2$; (d) $\frac{3}{5} + \frac{2}{25} = 0.68$.

17. (a) $0.714 < 5/7 < 0.715$; (b) $\sqrt{2} \geq 1.41$; (c) $7.32 \leq \sqrt{3} \leq 7.322$.

18. 0.526×10^{-4}; 0.429×10^9.

19. (a) If $a > \sqrt{n}$, then $b > \sqrt{n}$ also, so that $ab > n$, which is impossible. So the smaller factor a must be at most $\sqrt{n}$. (Note that if $a = \sqrt{n}$, then $b = \sqrt{n}$ also, and so $n = a^2$.)

(b) If a is prime, then a is the required prime factor of n; otherwise, a is composite. In this case a has a prime factor p such that $p < a$. Then p is a prime factor of n such that $p \leq \sqrt{n}$.

(c) If 89 is composite, then it must have a prime factor p such that $p \leq \sqrt{89}$. Since $9 < \sqrt{89} < 10$, we test 89 for divisibility by each prime less than 10, that is, by 2,3,5 and 7. Since none of these divides 89, we know 89 is prime.

(d) First note that $29 < \sqrt{899} < 30$. Hence if 899 is composite, it is divisible by one of the primes $2, 3, 5, ..., 29$. You will find one of these primes is a factor of 899, so that 899 is composite.

20. (a) 12. (b) 6.

(c) Yes, they each have exactly 6 digits in the recurring block. In the case of 1/13, 3/13 and 4/13 these digits are the same, but each fraction starts at a different point of the sequence.

(d) One method is to use the fact that the fraction $a/13$ has the same digits as $1/13$ in its recurring block if a is one of the sequence of remainders obtained when $1.000\ldots$ is divided by 13.

Exercises 2

1. (a) $\{1, 2, 3, 4, 5\}$; (b) $\{-1, 0, 1, 2, 3, 4\}$; (c) $\{3, 5, 7, 9\}$; (d) $\{10, 12, 14, 16, 18, 20\}$; (e) $\{1, 2, 4, 8, 16, 32\}$.

2. (a) $\{r : r \in \mathbb{Z} \text{ and } 12 \leq r \leq 17\}$; (b) $\{5m : m \in \mathbb{Z}\}$; (c) $\{-a : a \in \mathbb{Z} \text{ and } 1 \leq a \leq 10\}$; (d) $\{2n : n \in \mathbb{Z} \text{ and } 3 \leq n \leq 9\}$; (e) $\{5^k : k \in \mathbb{Z} \text{ and } k \geq 0\}$; (f) $\{(0.1)^s : s \in \mathbb{Z}^+\}$.

3. They are all equal; 3.

4. 10110; 01000; 11111; 00000; $\{a, u\}$.

5. $A \subseteq S$; $3 \in S$; $0 \in S$; $\emptyset \subseteq S$; $S \in \mathcal{P}(S)$; $A \in \mathcal{P}(S)$.

6. $A \cup B = \{1, 2, 4, 5, 6\}$; $A \cap B = \{4, 5\}$; $(A \cup B)' = \{0, 3\}$; $A - B = \{2\}$; $B - A = \{1, 6\}$; $A \oplus B = \{1, 2, 6\}$.

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P. Heidtmann and B. Jackson

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